Tutorial 6 - Solutions

1. (a) Note for this setting we are given that the population is normally distributed and the population variance is known and hence we use the test statistic

$$Z(\underline{X}) = \frac{\bar{X} - 50}{\sqrt{900/25}}.$$

When the null hypothesis is true we have $Z(\underline{X}) \sim \mathrm{N}(0,1)$. Therefore we construct a critical region by using the quantiles of a normal distribution. From the statistical tables we have

$$z_{0.05} = 1.6449$$

and hence the critical region in terms of Z is $(1.6449, \infty)$. Observe that we have a one-tailed test because of the form of the alternative hypothesis. Now we reject the null hypothesis if

$$Z(\underline{X}) = \frac{\bar{X} - 50}{\sqrt{900/25}} \geq 1.6449.$$

Rearranging we reject if

$$\bar{X} \geq 50 + 1.6449\sqrt{900/25} = 59.896.$$

(b) For the type II error we have

$$\begin{aligned} \beta &= \mathbb{P}(\text{do not reject } H_0 | H_1 \text{ is true}) = \mathbb{P}(\bar{X} < 59.9 | \mu = 56) \\ &= \mathbb{P}\left(Z < \frac{59.9 - 56}{30/5}\right) \quad \text{where } Z \sim \mathrm{N}(0,1) \\ &= \mathbb{P}(Z < 0.65) = 0.7422. \end{aligned}$$

2. With $X \sim \mathrm{Poi}(\lambda)$ we have:

(a) Probability of type I error is given by

$$\begin{aligned} \alpha &= \mathbb{P}(\text{reject } H_0 | H_0 \text{ is true}) = \mathbb{P}(X \geq 5 | \lambda = 2) \\ &= 1 - \mathbb{P}(X \leq 4 | \lambda = 2)) = 1 - 0.9473 = 0.0527 \quad \text{(use tables)} \end{aligned}$$

and for type II error

$$\beta = \mathbb{P}(\text{do not reject } H_0 | H_1 \text{ is true}) = \mathbb{P}(X \leq 4 | \lambda = 3) = 0.8153.$$

(b) The power of the test is

$$\gamma = \mathbb{P}(\text{reject } H_0 | H_1 \text{ is true}) = 1 - \beta = 0.1847.$$

3. (a) We have $n = 19$, $\sum x = 3676$ and $\sum x^2 = 728234$, giving $\bar{x} = 193.5$ and $s^2 = 945.8$. Calculating the CI as usual we get

$$\bar{x} \pm t_{18,0.025}\frac{s}{\sqrt{n}} = 193.5 \pm 14.82 = (178.68, 208.32).$$

Here we have assumed a normal population, which can be checked with a suitable plot of the data.

(b) The t-statistic is

$$\sqrt{19}(\bar{x} - 200)/s = -0.92.$$

The critical values at the 5% significance level are $t_{18,0.975} = -2.1$ and 2.1 (notice that this is a 2-sided test!). Therefore we *do not reject* the null hypothesis that $\mu = 200$ at 5% level of significance.

This agrees with the result obtained with the 95% CI, where the hypothesised value $\mu = 200$ was included in the interval. So the decision is consistent with the rule "reject if the CI does not contain the value specified by H_0".

(c) The p-value is (again, notice we have a 2-sided test)

$$\mathbb{P}(|t_{18}| > 0.92) = 2\mathbb{P}(t_{18} < -0.92) = 0.37.$$

This means that under H_0 the probability to obtain a value of the statistic as or more extreme as the observed value is 0.37. Obviously, this gives no evidence for rejecting H_0.

(d) The χ^2-statistic is

$$18s^2/2500 = 6.81.$$

Under H_0 this should be an observation from χ^2_{18} and we reject H_0 (at 1% significance level) if $\chi^2 < \chi^2_{18;0.99}$ (this is now a 1-sided test). Using statistical tables we get $\chi^2_{18;0.99} = 7.015$ and therefore we reject H_0 that $\sigma^2 = 2500$ in favour of $H_1 : \sigma^2 < 2500$ at the 1% level of significance.

The p-value is given by

$$p\text{-value} = \mathbb{P}(\chi^2_{18} < 6.81)$$

and using statistical tables (with linear interpolation) we get p-value = 0.0087. (In `R` we use `pchisq(6.81,18)` and obtain the more accurate p-value = 0.0084.)

Now the probability to observe the value of the statistic under H_0 is low and therefore we have evidence against H_0.

Exam Style Questions

(a) A type I error occurs if we reject H_0 when it is true, i.e. we find that 7 or more bowls are correctly identified when $p = 0.5$.

Let X denote the number of correctly identified bowls then $X \sim \text{Bin}(10, p)$ and

$$\begin{aligned}\mathbb{P}(\text{Type I error}) &= \mathbb{P}(X \geq 7; p = 0.5) \\ &= 1 - \mathbb{P}(X \leq 6; p = 0.5) \\ &= 1 - 0.8281 = 0.1719\end{aligned}$$

Here we have used the statistical tables for the binomial distribution.

[3 marks]

(b) (i) A type II error occurs if we fail to reject H_0 when H_1 is true. If H_1 is $p = 0.7$ then a type II error occurs if $X \leq 6$ when $p = 0.7$ and the probability of this is:

$$\mathbb{P}(X \leq 6; p = 0.7) = 0.3504$$

[2 marks]

(ii) If H_1 is $p = 0.8$ then a type II error occurs if $X \leq 6$ when $p = 0.8$ and the probability of this is:

$$\mathbb{P}(X \leq 6; p = 0.8) = 0.1209$$

[2 marks]

(c) The power of a test is the probability of rejecting H_0 when H_1 is true.

(i) For $H_1 : p = 0.7$ the power is $1 - 0.3504 = 0.6496$.

(ii) For $H_1 : p = 0.8$ the power is $1 - 0.1209 = 0.8791$.

This indicates that the test is more powerful as p increases.

[3 marks]